

Eliciting Care in VR: A Multi-Modal Pseudo-Haptic Approach to Virtual Object Fragility

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ABSTRACT

Conveying nuanced physical properties like fragility in Virtual Reality (VR) without specialized haptic hardware is a significant challenge [7]. This paper evaluates a multi-modal pseudo-haptic system that layers auditory feedback, a diegetic UI, and a procedural visual shader. Through a study ($N = 31$), we demonstrate that calibrated visual and auditory feedback can enable nuanced perceptual judgments and foster cautious behaviors equivalent to handling real-world delicate materials. Our findings suggest that layered feedback effectively induces a sense of "care" and psychological tension, enriching interaction plausibility in consumer VR.

Index Terms: Pseudo-Haptics, VR, Fragility, Hand Tracking, Multi-modal Feedback, Ablation Study.

1 INTRODUCTION

Current consumer VR hardware excels at visual immersion but lacks the fidelity to convey physical properties like **fragility** [7]. While high-fidelity haptic devices like force-feedback exoskeletons exist, they remain prohibitively expensive for widespread adoption. This motivates software-driven techniques like **pseudo-haptics**, which exploit fundamental perceptual mechanisms like *visual dominance*—the brain's tendency to prioritize visual information over conflicting sensory inputs—to create compelling physical illusions [5].

Fragility is not just a physical threshold; it is an affective property that inherently elicits emotional responses, such as "care" or "tension" [3]. We propose a software-only approach to induce **sensory recalibration** [1] by creating a deliberate mismatch between actions and feedback.

2 SYSTEM DESIGN

The system maps grip intensity $G \in [0, 1]$ from controller-less hand tracking on a Meta Quest 3. Our design focuses on a closed-loop model where grip intensity influences the object's state in real-time through multi-modal cues.

2.1 Multi-Modal Feedback Layers

1. Procedural Stress Shader: As G increases, a shader dynamically calculates a Voronoi-based crack pattern. Visual deformation is a dominant cue for material perception [6]. Crack width W is defined as:

$$W = \max \left(0, \frac{G - T_{\text{warn}}}{T_{\text{break}} - T_{\text{warn}}} \right) \quad (1)$$

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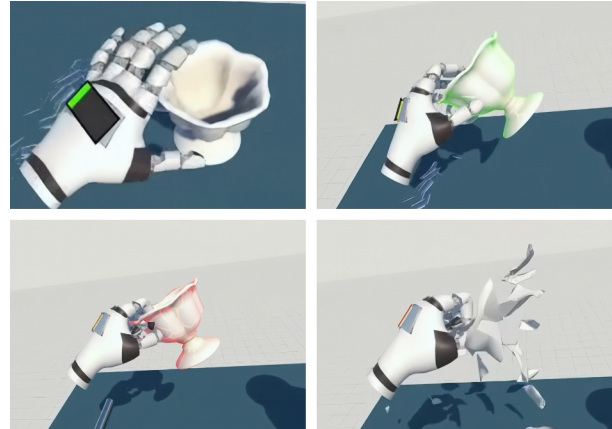


Figure 1: Visual Feedback: Real-time procedural cracking on a virtual object during a high-intensity grasp.

where T_{warn} is the threshold for visible damage (fig. 1).

2. Auditory Synthesis: Audio acts as a proxy for tactile micro-fractures, which is especially effective in visual-haptic conflicts [4]. Volume (S_{vol}) and pitch (S_{pitch}) are modulated based on the grip's first derivative:

$$S_{\text{vol}} = \alpha \cdot \log \left(1 + \left| \frac{dG}{dt} \right| \right), \quad S_{\text{pitch}} = \beta \cdot G \quad (2)$$

This logarithmic mapping mimics human perception of loudness, providing a "visceral" warning when the user squeezes too quickly [9].

3. Embodiment and Agency: For pseudo-haptic illusions to be effective, the user must feel ownership over their virtual hands, known as the *Virtual Hand Illusion* (VHI) [2, 8]. We represent hands as **robotic avatars** to mitigate the "uncanny valley" effect—where photorealistic human hands without tactile feedback create sensory dissonance—while maintaining high agency scores.

3 USER STUDY AND RESULTS

We recruited 31 participants age ($M = 24.2$, $SD = 4.1$). To evaluate robustness, we employed a mixed design: an **Incremental Group** ($N = 21$) experienced additive feedback, while an **Ablation Group** ($N = 10$) experienced modalities in isolation. We acknowledge the sample size disparity as a current weakness and plan to balance these groups in future research. Participants interacted with three objects of increasing fragility: a Vase (Low), Teacup (Medium), and Test Tube (High).

3.1 Perceptual Discrimination

A Friedman test revealed significant differences across objects for Perceived Fragility ($\chi^2(2) = 50.47, p < .001$), Sense of Care ($\chi^2(2) = 49.23, p < .001$), and Tension ($\chi^2(2) = 42.22, p < .001$).

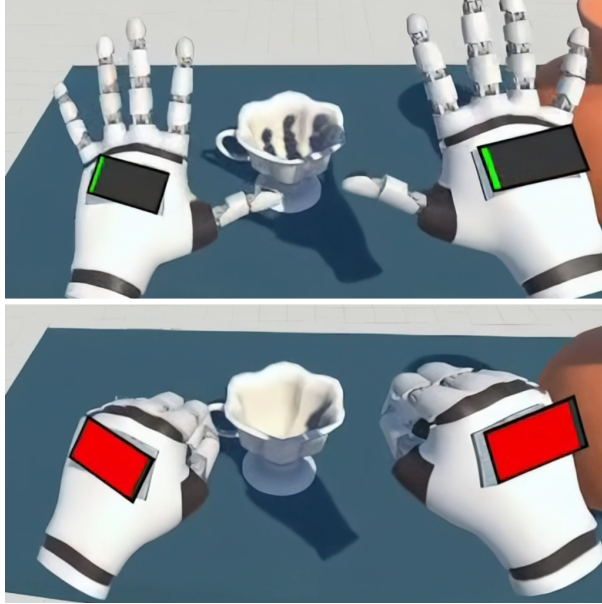


Figure 2: Diegetic UI: The robotic hand integrates a safety gauge providing real-time feedback on the grip force G .

Table 1: Mean (SD) Likert Ratings (1 – 7) for the sample ($N = 31$).

Object	Fragility	Care	Tension
Vase	2.40 (1.69)	1.93 (1.44)	1.93 (1.53)
Teacup	4.47 (1.43)	3.93 (1.62)	3.30 (1.68)
Test Tube	6.63 (0.76)	6.47 (0.94)	5.80 (1.52)

3.2 Behavioral Impact and Robustness

The system induced a **40% reduction in hand velocity** when handling the Test Tube. Mann-Whitney U tests confirmed **no significant difference** in final ratings between the Incremental and Ablation groups ($p > .7$), suggesting that the perception of fragility is robust and independent of the learning path. Qualitative feedback described a transition from initial anxiety to a "therapeutic" release when intentionally breaking objects, indicating a profound sense of presence and agency.

4 CONCLUSION

Our system effectively conveys fragility through multi-modal integration, inducing psychological tension that substitutes for missing haptic resistance. Future work will analyze kinematic data, such as jerk, to further quantify this careful behavior.

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REFERENCES

[1] <https://doi.org/10.1109/3DUI.2016.7460049>F. Argelaguet, C. Andujar, and A. Lécuyer. <https://doi.org/10.1109/3DUI.2016.7460049>Pseudo-

haptic feedback for tangible interaction. <https://doi.org/10.1109/3DUI.2016.7460049>In *Proc. IEEE Symposium on 3D User Interfaces (3DUI)*, <https://doi.org/10.1109/3DUI.2016.7460049pp>. 169–172, <https://doi.org/10.1109/3DUI.2016.74600492016>. <https://doi.org/10.1109/3DUI.2016.7460049> doi: 10.1109/3DUI.2016.7460049

[2] <https://doi.org/10.3389/fpsyg.2023.1119561>A. M. de Carvalho Souza, R. Barrocas, M. H. Fischer, E. Arnaud, K. Moeller, and C. Rennó-Costa. <https://doi.org/10.3389/fpsyg.2023.1119561>Combining virtual reality and tactile stimulation to investigate embodied finger-based numerical representations. <https://doi.org/10.3389/fpsyg.2023.1119561>*Frontiers in Psychology*, <https://doi.org/10.3389/fpsyg.2023.111956114>, <https://doi.org/10.3389/fpsyg.2023.11195612023>. <https://doi.org/10.3389/fpsyg.2023.1119561> doi: 10.3389/fpsyg.2023.1119561

[3] <https://doi.org/10.1109/TOH.2021.3097143J>. DiDonato, F. Chinello, M. A. Otaduy, V. Pettarin, and S. Serafin. <https://doi.org/10.1109/TOH.2021.3097143>Perception of material properties from audio-visual feedback of tool-surface interaction. <https://doi.org/10.1109/TOH.2021.3097143>*IEEE Transactions on Haptics*, [https://doi.org/10.1109/TOH.2021.3097143\(3\):671–682](https://doi.org/10.1109/TOH.2021.3097143(3):671-682), <https://doi.org/10.1109/TOH.2021.30971432021>. <https://doi.org/10.1109/TOH.2021.3097143> doi: 10.1109/TOH.2021.3097143

[4] S. Guerraz, J. B. Toinon, M. Mposa, N. Hatsopoulos, and K. Giraud. Auditory feedback and visual-haptic conflict in a virtual environment. In *Proc. 5th International Conference on Enactive Interfaces*, pp. 1–4, 2008.

[5] <https://doi.org/10.1162/pres.18.1.39A>. Lécuyer. <https://doi.org/10.1162/pres.18.1.39>Simulating haptic feedback using vision: A survey of the state-of-the-art of pseudo-haptic feedback. <https://doi.org/10.1162/pres.18.1.39>*Presence: Teleoperators and Virtual Environments*, [https://doi.org/10.1162/pres.18.1.3918\(1\):39–53](https://doi.org/10.1162/pres.18.1.3918(1):39-53), <https://doi.org/10.1162/pres.18.1.392009>. <https://doi.org/10.1162/pres.18.1.39> doi: 10.1162/pres.18.1.39

[6] <https://doi.org/10.1109/WHC.2017.7989871C>. Paul, S. Degallier, M. Fiorentino, P. Cher-Micheas, S. Haliyo, and S. Regnier. <https://doi.org/10.1109/WHC.2017.7989871>Visual dominance and texture perception for virtual material rendering. <https://doi.org/10.1109/WHC.2017.7989871>In *Proc. IEEE World Haptics Conference (WHC)*, <https://doi.org/10.1109/WHC.2017.7989871pp>. 25–30, <https://doi.org/10.1109/WHC.2017.79898712017>. <https://doi.org/10.1109/WHC.2017.7989871> doi: 10.1109/WHC.2017.7989871

[7] <https://doi.org/10.1145/3055304R>. Skarbez, F. P. Brooks Jr, and M. C. Whitton. <https://doi.org/10.1145/3055304>A survey of presence and related concepts. <https://doi.org/10.1145/3055304>*ACM Computing Surveys (CSUR)*, [https://doi.org/10.1145/305530450\(6\):1–39](https://doi.org/10.1145/305530450(6):1-39), <https://doi.org/10.1145/30553042017>. <https://doi.org/10.1145/3055304> doi: 10.1145/3055304

[8] <https://doi.org/10.3389/neuro.01.029.2009M>. Slater, D. Perez-Marcos, H. H. Ehrsson, and M. V. Sanchez-Vives. <https://doi.org/10.3389/neuro.01.029.2009>Inducing illusory ownership of a virtual body. <https://doi.org/10.3389/neuro.01.029.2009>*Frontiers in Neuroscience*, <https://doi.org/10.3389/neuro.01.029.20093:214>, <https://doi.org/10.3389/neuro.01.029.20092009>. <https://doi.org/10.3389/neuro.01.029.2009> doi: 10.3389/neuro.01.029.2009

[9] <https://doi.org/10.3389/fpsyg.2019.03001T>. R. Stanton and C. Spence. <https://doi.org/10.3389/fpsyg.2019.03001>The influence of auditory cues on bodily and movement perception. <https://doi.org/10.3389/fpsyg.2019.03001>*Frontiers in Psychology*, <https://doi.org/10.3389/fpsyg.2019.03001110>, <https://doi.org/10.3389/fpsyg.2019.030012020>. <https://doi.org/10.3389/fpsyg.2019.03001> doi: 10.3389/fpsyg.2019.03001